



Project Proposal

AmarCure

An AI-Powered Comprehensive Healthcare Management System

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1. Executive Summary :

Managing personal healthcare records in Bangladesh remains a major challenge due to fragmented, paper-based systems that make medical information difficult to access, organize, and interpret. “AmarCure” is an AI-powered healthcare management platform built to transform this experience by digitizing and centralizing personal health records in one secure, intelligent ecosystem.

Beyond secure document storage and automated appointment scheduling, AmarCure leverages artificial intelligence to analyze complex medical reports and translate them into clear, patient-friendly insights. By making healthcare information more accessible and actionable, AmarCure empowers individuals and families to take a proactive role in monitoring their health, making informed decisions, and achieving better long-term health outcomes.

2. Problem Statement :

The healthcare system in Bangladesh faces several challenges that make it difficult for patients and healthcare providers to manage medical information efficiently -

- **Paper-Based Medical Records:** Most hospitals and clinics still use paper reports and prescriptions. These records can be easily lost, damaged, or misplaced, making it difficult for doctors to access a patient's complete medical history.
- **Lack of Emergency Medical Information:** In emergencies, important information such as blood group, allergies, current medications, and existing diseases is often unavailable. This can delay proper treatment and increase health risks.
- **Poor Medication Management:** Many patients, especially elderly people and those with chronic diseases, forget to take their medicines on time or miss follow-up appointments, which can worsen their health.
- **Difficulty in Family Health Monitoring:** Family members often find it difficult to monitor the health of their loved ones because medical records are kept in different hospitals, clinics, or at home.

- **Limited Understanding of Medical Reports:** Most patients cannot easily understand medical reports or laboratory test results. They often do not know whether important health indicators, such as blood sugar or cholesterol levels, are improving or getting worse.
- **Limited Features in Existing Healthcare Apps:** Most current healthcare applications only provide services like online doctor appointments or telemedicine. They do not offer a complete solution for storing medical records, tracking health history, or providing AI-based health insights.
- **Inefficient Healthcare Management:** Because there is no single platform to store and manage health records, patients often repeat the same medical tests at different hospitals. This increases healthcare costs, wastes time, and delays proper treatment.

3. Proposed Solution: AmarCure :

To resolve these challenges, AmarCure offers a centralized platform that safely stores health data while making it actionable and understandable for the average user.

1. Centralized Medical Records :

Users can securely upload and store prescriptions, laboratory reports, X-rays, MRI/CT scan reports, vaccination records, discharge summaries, and other medical documents in one secure location.

2. AI-Powered Medical Report Analysis :

The platform uses AI to explain complex medical reports in simple language. It identifies whether important health indicators such as blood sugar, cholesterol, hemoglobin, kidney function, or liver function are within the normal range and provides an easy-to-understand summary.

4. Medicine Reminder System :

Users receive notifications reminding them to take medications at the correct time. The system also tracks missed doses and medication completion history.

5. Appointment & Follow-up Reminder :

The application reminds users about upcoming doctor appointments, follow-up visits, vaccinations, and scheduled laboratory tests.

6. Family & Caregiver Dashboard :

Users can securely share access with trusted family members or caregivers. They can monitor medication schedules, appointments, and important health updates, which is especially helpful for elderly patients and children.

7. Emergency Health Profile :

A one-click emergency profile displays critical information including:

- Blood group
- Allergies
- Chronic diseases
- Current medications
- Emergency contacts
- Medical conditions

This information can be accessed quickly during emergencies.

8. Personal Health Timeline :

All medical activities, including doctor visits, prescriptions, diagnoses, laboratory reports, and treatments, are organized in a chronological timeline for easy tracking.

9. Smart Health Dashboard :

The dashboard provides a quick overview of:

- Recent reports
- Upcoming appointments
- Medicine schedule
- Pending medical tasks

10. OCR-Based Document Scanning :

Users can simply take a picture of a prescription or medical report. Optical Character Recognition (OCR) automatically extracts the important information and stores it digitally.

11. Intelligent Search :

Users can instantly search medical records by:

- Doctor name
- Hospital
- Disease
- Date
- Medicine name

12. Doctor Visit Notes :

Users can save doctor's advice, symptoms, treatment plans, and personal observations after every consultation.

13. Health Profile Management :

The platform stores important personal health information, including:

- Blood group
- Height and weight
- Allergies
- Chronic illnesses
- Vaccination history
- Emergency contacts

14. Health Risk Alerts :

If uploaded reports contain abnormal values, the AI alerts users and recommends consulting a healthcare professional. This feature is intended for awareness only and does not replace medical advice.

15. Medicine History :

Users can view their previous medications, treatment history, and prescription records, making it easier to share accurate information with doctors.

16. Multi-User Support :

A single family account can manage multiple health profiles for parents, children, or elderly family members while keeping each person's records separate and secure.

17. Health Document Backup :

Medical records are automatically backed up to the cloud, reducing the risk of losing important healthcare documents.

18. Download & Share Reports :

Users can download, print, or securely share selected medical documents with doctors, hospitals, or family members whenever required.

19. Personalized Health Tips :

Based on the user's stored health information and AI analysis, the system provides simple lifestyle recommendations such as staying hydrated, maintaining a balanced diet, exercising regularly, or scheduling routine health checkups. These suggestions are general wellness guidance and not medical advice.

20. Future Scope :

Future versions of AmarCure may include:

- Online doctor appointments.
- Telemedicine support.
- Integration with wearable devices (smartwatches and fitness bands)
- Pharmacy integration for medicine ordering.
- Health insurance record management.
- AI chatbot for answering common health-related questions.
- Integration with national digital health systems.

4. Technical Requirements:

AmarCure uses modern technologies to make the platform secure, fast, and easy to use.

- **Frontend (React):**

React is used to create a simple and user-friendly interface. It helps users easily view and manage their health information.

- **Backend (Node.js):**

Node.js manages the system's main operations, such as processing data, handling user requests, and sending reminders.

- **Database & Security (Supabase):**

Supabase stores user health data securely and provides login authentication to protect personal information.

5. Conclusion :

In conclusion, AmarCure helps solve the problems of traditional paper-based healthcare records by providing a simple and smart digital healthcare solution. It keeps medical information organized, secure, and easily accessible while helping users better understand their health. With features like AI support, medical record management, and health reminders, AmarCure can improve healthcare management and make it more convenient for people in Bangladesh.